



Python Programming - 2301CS404

Lab - 8

Roll No : 418

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01) WAP to find their union, intersection, and difference of the given two sets.

Sample Input:

A = {1, 2, 3, 4}
B = {3, 4, 5, 6}

Sample Output:

Union: {1, 2, 3, 4, 5, 6}
Intersection: {3, 4}
Difference (A-B): {1, 2}

```
In [5]: A = {1, 2, 3, 4}
        B = {3, 4, 5, 6}
        Union = A | B
        Intersection = A & B
        Difference = A - B
        print("Union : ", Union)
        print("Intersection : ", Intersection)
        print("Difference (A-B): ", Difference)
```

Union : {1, 2, 3, 4, 5, 6}
Intersection : {3, 4}
Difference (A-B): {1, 2}

02) WAP to Find elements common to all three given sets.

Sample Input:

```
A = {1, 2, 3, 4}
B = {2, 3, 5}
C = {2, 3, 6}
```

Sample Output:

```
{2, 3}
```

```
In [7]: A = {1, 2, 3, 4}
        B = {2, 3, 5}
        C = {2, 3, 6}
        res = A & B & C
        print("Common Element : ", res)
```

```
Common Element : {2, 3}
```

03) WAP to remove duplicate elements from a list while preserving unique values.**Sample Input:**

```
[10, 20, 10, 30, 20, 40]
```

Sample Output:

```
{10, 20, 30, 40}
```

```
In [4]: lt = [10, 20, 10, 30, 20, 40]
        unique = set(lt)
        print(unique)

        # Extra....
        # lst = [10, 20, 10, 30, 20, 40]
        # # Using dict.fromkeys to remove duplicates while preserving order
        # unique_dict = dict.fromkeys(lst)
        # print(unique_dict)
```

```
{40, 10, 20, 30}
```

04) WAP to count how many elements are common between two sets.**Sample Input:**

```
A = {1, 2, 3, 4}
B = {3, 4, 5, 6}
```

Sample Output:

```
2
```

```
In [5]: A = {1, 2, 3, 4}
B = {3, 4, 5, 6}
res = A & B
count = len(res)
print("Common Between Two Sets : ", count)

# Using intersection() method
# common_count = len(A.intersection(B))
# print(common_count)

# Using a Loop
# common_count = sum(1 for x in A if x in B)
# print(common_count)
```

Common Between Two Sets : 2

2

05) WAP to Merge two dictionaries. If a key exists in both dictionaries, add their values.

Sample Input:

```
Dict1 = {1: 10, 2: 20, 3: 30}
Dict2 = {2: 40, 3: 50, 4: 60}
```

Sample Output:

```
{1: 10, 2: 60, 3: 80, 4: 60}
```

```
In [6]: Dict1 = {1: 10, 2: 20, 3: 30}
Dict2 = {2: 40, 3: 50, 4: 60}
result = Dict1.copy()
for key, value in Dict2.items():
    if key in result:
        result[key] += value
    else:
        result[key] = value
print(result)
```

```
{1: 10, 2: 60, 3: 80, 4: 60}
```

06) WAP to Sort a dictionary in ascending order of its values.

Hint: in sorted(), use udf for key parameter

Sample Input:

```
{'Math': 70, 'Physics': 85, 'Chemistry': 60}
```

Sample Output:

```
{'Chemistry': 60, 'Math': 70, 'Physics': 85}
```

```
In [59]: import operator
d1 = {'Math': 70, 'Physics': 85, 'Chemistry': 60}
sorted_items = sorted(d1.items(), key=operator.itemgetter(1))
ans = dict(sorted_items)
print("Original Dictionary:", d1)
print("Sorted Dictionary (by value):", ans)
```

Original Dictionary: {'Math': 70, 'Physics': 85, 'Chemistry': 60}

Sorted Dictionary (by value): {'Chemistry': 60, 'Math': 70, 'Physics': 85}

07) WAP to find the keys having maximum and minimum values in a dictionary.

Sample Input:

```
{'A': 50, 'B': 90, 'C': 30}
```

Sample Output:

Max Value Key: B

Min Value Key: C

```
In [9]: # dt = {'A': 50, 'B': 90, 'C': 30}
# max1 = dt['A']
# min1 = dt['B']
# for i, j in dt.items():
#     if max1 < j:
#         max1 = j
#         maxEle = i
#     if min1 > j:
#         min1 = j
#         minEle = i
# print("Max Value Key: ", maxEle)
# print("Min Value Key: ", minEle)

# # Sample input
# marks = {'A': 50, 'B': 90, 'C': 30}
# items_list = list(marks.items())
# max_key, max_value = items_list[0]
# min_key, min_value = items_list[0]

# # Find max and min values and their keys
# for key, value in marks.items():
#     if value > max_value:
#         max_value = value
#         max_key = key
#     if value < min_value:
#         min_value = value
#         min_key = key

# print(f"Max Value Key: {max_key}")
# print(f"Min Value Key: {min_key}")
```

```
dt = {'A': 50, 'B': 90, 'C': 30}
maxEle = max(dt, key = dt.get)
minEle = min(dt, key = dt.get)
print("Max Value Key: ", maxEle)
print("Min Value Key: ", minEle)
```

Max Value Key: B

Min Value Key: C

08) WAP to create a dictionary with numbers from 1 to N as keys and their squares as values.

Sample Input:

N = 5

Sample Output:

{1: 1, 2: 4, 3: 9, 4: 16, 5: 25}

```
In [9]: d1 = dict(((i, i**2) for i in range(1, 6)))
print(d1)
```

```
# Sample input
# N = 5
# result = {i: i*i for i in range(1, N+1)}
# print(result)
```

```
#Sample input
# N = 5
# result = {}
# for i in range(1, N+1):
#     result[i] = i * i
# print(result)
```

{1: 1, 2: 4, 3: 9, 4: 16, 5: 25}

09) WAP to count the frequency of each character in a string using a dictionary.

Sample Input:

programming

Sample Output:

{'p': 1, 'r': 2, 'o': 1, 'g': 2, 'a': 1, 'm': 2, 'i': 1, 'n': 1}

```
In [11]: str1 = "programming"
res = {}
for i in str1:
    if i not in res:
        res[i] = 1
    else:
```

```

        res[i] += 1
    print(res)

    # using get() method:
    # string = "programming"
    # freq = {}
    # for char in string:
    #     freq[char] = freq.get(char, 0) + 1
    # print(freq)

```

```
{'p': 1, 'r': 2, 'o': 1, 'g': 2, 'a': 1, 'm': 2, 'i': 1, 'n': 1}
```

10) WAP to swap keys and values of a dictionary.

Sample Input:

```
{1: 'A', 2: 'B', 3: 'C'}
```

Sample Output:

```
{'A': 1, 'B': 2, 'C': 3}
```

```

In [13]: # d1 = {1: 'A', 2: 'B', 3: 'C'}
# inv_di = {}
# for k,v in d1.items():
#     inv_di[v] = k
# print(inv_di)

original_dict = {1: 'A', 2: 'B', 3: 'C'}
swapped_dict = {value: key for key, value in original_dict.items()}
print(swapped_dict)

```

```
{'A': 1, 'B': 2, 'C': 3}
```

11) WAP to find the student with the highest total marks, given student names and their marks.

Sample Input:

```
{'Amit': [70, 80, 90], 'Riya': [85, 75, 80], 'Neha': [60, 70, 65]}
```

Sample Output:

```
Topper: Amit
```

```

In [14]: d1 = {'Amit': [70, 80, 90], 'Riya': [85, 75, 80], 'Neha': [60, 70, 65]}
highest = 0
toppername = ""
for i, j in d1.items():
    s = 0
    s += sum(j)
    if highest < s:
        highest = s
        toppername = i
print("Highest Marks : ", highest)

```

```

print("Topper Name: ", toppername)

# Sample input
# students = {'Amit': [70, 80, 90], 'Riya': [85, 75, 80], 'Neha': [60, 70, 65]}
# max_total = -1
# topper = ""
# for student, marks in students.items():
#     total = sum(marks)
#     if total > max_total:
#         max_total = total
#         topper = student
# print(f"Topper: {topper}")

```

Highest Marks : 240

Topper Name: Amit

Topper: Amit

12) WAP to extract keys whose values are even numbers.

Sample Input:

```
{1: 11, 2: 20, 3: 15, 4: 40}
```

Sample Output:

```
[2, 4]
```

```

In [21]: d1 = {1: 11, 2: 20, 3: 15, 4: 40}
res = list(i for i in d1 if i % 2 == 0)
print("even Key : ", res)

# Sample input
# num_dict = {1: 11, 2: 20, 3: 15, 4: 40}
# even_keys = [key for key, value in num_dict.items() if value % 2 == 0]
# print(even_keys)

# Alternative without list comprehension:
# num_dict = {1: 11, 2: 20, 3: 15, 4: 40}
# even_keys = []
# for key, value in num_dict.items():
#     if value % 2 == 0:
#         even_keys.append(key)
# print(even_keys)

```

even Key : [2, 4]

13) WAP to Find common keys between two dictionaries.

Sample Input:

```
Dict1 = {1: 'A', 2: 'B', 3: 'C'}
Dict2 = {2: 'X', 3: 'Y', 4: 'Z'}
```

Sample Output:

[2, 3]

```
In [25]: Dict1 = {1: 'A', 2: 'B', 3: 'C'}
Dict2 = {2: 'X', 3: 'Y', 4: 'Z'}
common_keys = Dict1.keys() & Dict2.keys()
print("Dictionary 1:", Dict1)
print("Dictionary 2:", Dict2)
print("Common keys:", list(common_keys))

# Sample input
# Dict1 = {1: 'A', 2: 'B', 3: 'C'}
# Dict2 = {2: 'X', 3: 'Y', 4: 'Z'}
# common_keys = [key for key in Dict1 if key in Dict2]
# print(common_keys)

# Alternative using set intersection:
# Dict1 = {1: 'A', 2: 'B', 3: 'C'}
# Dict2 = {2: 'X', 3: 'Y', 4: 'Z'}
# common_keys = list(set(Dict1.keys()) & set(Dict2.keys()))
# print(common_keys)
```

Dictionary 1: {1: 'A', 2: 'B', 3: 'C'}

Dictionary 2: {2: 'X', 3: 'Y', 4: 'Z'}

Common keys: [2, 3]

14) WAP to handle missing keys in dictionaries.

Sample Input - 1:

```
dict1 = {'a': 5, 'c': 8, 'e': 2}
key = 'c'
```

Sample Output - 1:

8

Sample Input - 2:

```
dict1 = {'a': 5, 'c': 8, 'e': 2}
key = 'd'
```

Sample Output - 2:

Key Not Found

```
In [31]: dict1 = {'a': 5, 'c': 8, 'e': 2}
key = input("Enter key: ")
if key in dict1:
    print(dict1[key])
else:
    print("Key Not Found")

# Using get() method:
# dict1 = {'a': 5, 'c': 8, 'e': 2}
# key = input("Enter key: ")
```



```
# value = dict1.get(key, "Key Not Found")  
# print(value)
```

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15) WAP to create a dictionary using dictionary comprehension from two lists containing student names and marks. Include only those students in the dictionary whose marks are greater than or equal to 75.

Sample Input:

```
students = ['Amit', 'Riya', 'Neha', 'Karan']  
marks = [78, 72, 85, 60]
```

Sample Output:

```
{'Amit': 78, 'Neha': 85}
```

```
In [42]: students = ['Amit', 'Riya', 'Neha', 'Karan']  
marks = [78, 72, 85, 60]  
dt = dict(zip(students, marks))  
res = {}  
for k, v in dt.items():  
    if dt[k] > 75:  
        res[k] = v  
print(res)  
  
# Alternative using zip():  
# students = ['Amit', 'Riya', 'Neha', 'Karan']  
# marks = [78, 72, 85, 60]  
# result = {name: mark for name, mark in zip(students, marks) if mark >= 75}  
# print(result)
```

```
{'Amit': 78, 'Neha': 85}
```

```
In [ ]:
```